University Of Barishal

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (CSE)

LAB FINAL EXAMINATION - 2026

3rd YEAR, 2nd SEMESTER (2022-23)

COURSE CODE: CSE-3212

COURSE TITLE: Mobile App Development Lab

Part-01 (Theory)

Time: 30 Minutes **Full Marks:** 15

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1. *What is state management, and what distinguishes local from global state?* **02**
2. *Explain how GetX reactive state management works and why it is memory efficient.* **02**
3. *Why is asynchronous programming (async/await) essential for network requests in Flutter?* **02**
4. **When will you use SingleChildScrollView?** **01**
5. *Explain the role of JSON deserialization and HTTP status codes in client-server communication.* **02**
6. **Data Types from JSON** **02**

For {"id":101,"elements":["Arif","Ariful",45.3,{"roll": "22CSE002"},"Arifa"]}:

a) Dart type of the whole object?

b) Dart type of "*elements*"?

7. *Why does separating UI from business logic improve code maintainability and testability?* **02**
8. Give one **technical** advantage of Native (Kotlin/Swift) and one **business** advantage of Flutter. **02**

Part-02 (App Development)

Time: 2 Hours **Full Marks: 45**

Develop an mobile app using any frameworks/language complying the following instructions , APIs & Figma Design:

→ Write Down 'Questions' & 'Category' Model in the answer-script

API Endpoints:

- Categories: https://opentdb.com/api_category.php
- Questions: https://opentdb.com/api.php?amount=<number>&category=<id>&difficulty=<level>&type=multiple

App's UI / Design prototype here: https://tinyurl.com/cse9-lab-exam-figma

Instructions: You don't need to follow the design 100%. Your goal is to get it as close to the design as possible. **Instructions link:** https://tinyurl.com/cse9-mad-lab-project-ins

Home → Categories → Start Quiz Form(Config) → Quiz → Results → Play Again.

Best of Luck

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